5 **(a)** Three resistors of resistances $R_1$, $R_2$ and $R_3$ are connected in a network in parallel as shown in Figure 5.1.

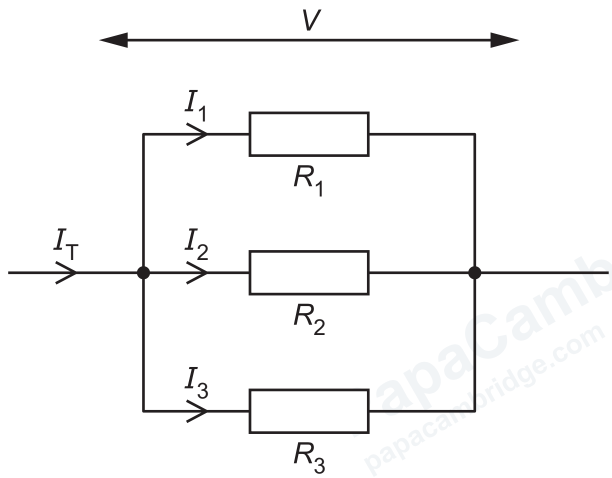


**Figure 5.1**

The currents in the resistors are $I_1$, $I_2$ and $I_3$ respectively. The total current entering the network is $I_T$. The potential difference (p.d.) across the resistors is $V$.

Derive the equation

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

where $R_T$ is the total resistance of the network.

[2]

**(b)** Two identical resistance wires, each of resistance 2.8 Ω, are connected in parallel to a battery with negligible internal resistance and electromotive force (e.m.f.) 12 V as shown in Figure 5.2.

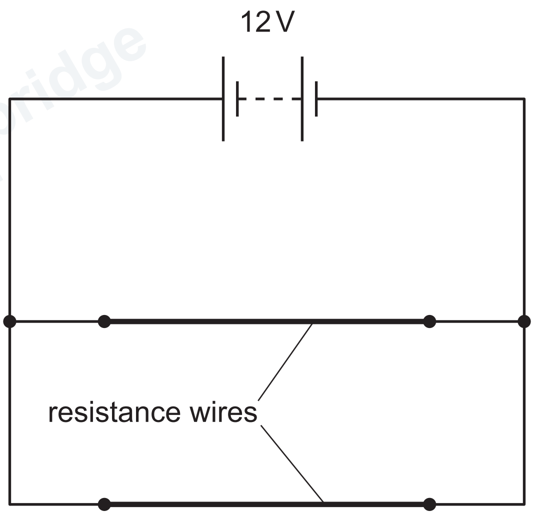


**Figure 5.2**

**(i)** Calculate the current in the battery.

current = ........................................................ A [2]

**(ii)** Each resistance wire has a cross-sectional area of $6.1 \times 10^{-6}\,m^2$.

The wires are made from a metal with a number density of charge carriers (free electrons) of $8.5 \times 10^{28}\,m^{-3}$.

Calculate the average drift speed of the charge carriers in the wires.

speed = ................................................... $m\,s^{-1}$ [2]

**(c)** A component X is added to the circuit in **5(b)** as shown in Figure 5.3.

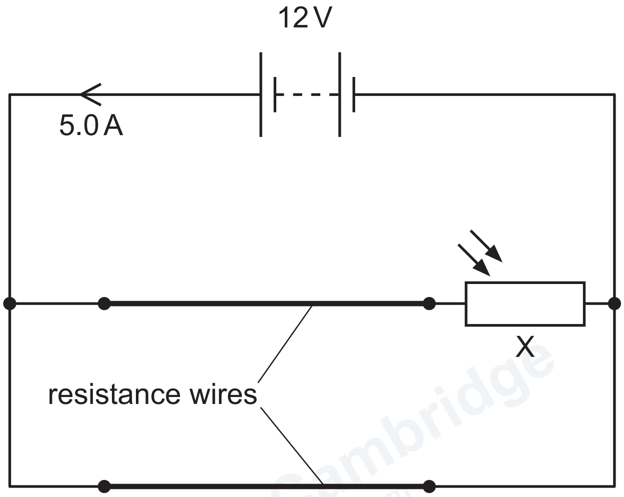


**Figure 5.3**

The current in the battery is now 5.0 A.
The resistance of each resistance wire remains 2.8 Ω.

**(i)** State the name of component X.

.............................................................................................................................. [1]

**(ii)** Determine the resistance of component X.

resistance = .................................................... Ω [3]

**(iii)** A change in the environment of component X causes the current in the battery to decrease. The resistance of the resistance wires remains constant.

State and explain how the environment of X has changed.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................. [2]

[Total: 12]